Title : IMDB Sentiment Classification Using LSTM

Aim

- Use IMDB sentiment classification dataset.
- Implement an LSTM for binary classification.
- Plot accuracy and visualize example predictions.

Theory

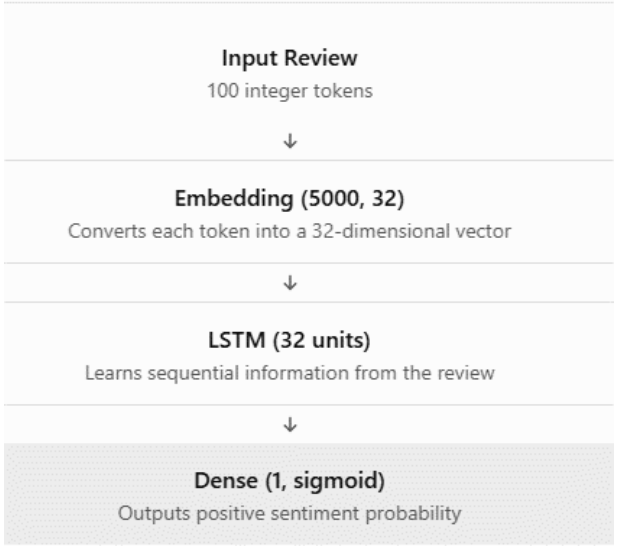
Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network (RNN) designed to learn sequential dependencies and retain relevant information over time. It is useful for processing text, speech, and time-series data.

In sentiment classification, LSTM analyzes a movie review and predicts whether its sentiment is positive or negative.

The IMDB dataset contains 50,000 movie reviews labeled as positive (1) or negative (0). In this experiment, the dataset is used to train and test an LSTM model.

The major components of the model are:

1. Embedding Layer: Converts integer-encoded words into dense numerical vectors.
2. LSTM Layer: Learns sequential patterns and contextual relationships between words.
3. Dense Layer: Uses the sigmoid activation function to predict the probability of positive sentiment.
4. Adam Optimizer: Updates model weights to minimize the loss function.
5. Binary Cross-Entropy: Measures the difference between predicted probabilities and actual sentiment labels.



The reviews are padded to a fixed length so that they can be processed efficiently in batches. The model is trained using the training dataset and evaluated using the test dataset.

The accuracy graph compares training and validation accuracy across epochs, helping assess the learning performance of the model.

The predicted sentiment is determined using the following rule:

- If the predicted probability is greater than or equal to 0.5, the review is classified as positive.
- If the predicted probability is less than 0.5, the review is classified as negative.

Algorithm

1. Start.
2. Import the required Python libraries.
3. Load the IMDB movie review dataset with a vocabulary size of 5,000 words.
4. Pad the training and testing review sequences to a maximum length of 100.
5. Construct an LSTM model consisting of an embedding layer, an LSTM layer, and a dense output layer with sigmoid activation.
6. Compile the model using the Adam optimizer, binary cross-entropy loss, and accuracy metric.
7. Train the model for three epochs using the training dataset and validate it using a portion of the training data.
8. Evaluate the trained model using the test dataset and display the test accuracy.
9. Plot the training and validation accuracy against the number of epochs.
10. Predict the sentiment of five test reviews.
11. Display the predicted and actual sentiment labels for each review.
12. Stop.

After implementing the LSTM model for IMDB sentiment classification, perform the following experiments to study how different parameters affect model performance.

Experimental Tasks

1. Change the number of epochs from 3 to 5 and 10.
2. Change the number of LSTM units from 32 to 64.
3. Change the batch size from 128 to 64.
4. Change the maximum review length from 100 to 200.
5. Compare the training accuracy, validation accuracy, and test accuracy for each experiment.

Change only one parameter at a time while keeping the other parameters constant.

This helps identify the effect of each parameter on model performance.

Observation Table

Exp. No.	Parameter Changed	Value	Training Accuracy (%)	Validation Accuracy (%)	Test Accuracy (%)
1	Epochs	3			
2	Epochs	5			
3	Epochs	10			
4	LSTM units	64			
5	Batch size	64			
6	Maximum review length	200			

Conclusion

The LSTM model was implemented successfully for binary sentiment classification using the IMDB movie review dataset. The model was trained to classify reviews as positive or negative. Its performance was evaluated using test accuracy, and a graph was plotted to compare training and validation accuracy. Example predictions were also compared with actual sentiment labels to understand the classification performance of the model.